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phylo2vec: a library for vector-based phylogenetic tree manipulation

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Summary

Phylogenetics is a fundamental component of evolutionary analysis frameworks in biology (Yang, 2014) and linguistics (Atkinson & Gray, 2005). Recently, the advent of large-scale genomics and the SARS-CoV-2 pandemic has highlighted the necessity for phylogenetic software to handle large datasets (Attwood et al., 2022; Kapli et al., 2020; Khurana et al., 2024; Kraemer et al., 2025). While significant efforts have focused on scaling optimisation algorithms (De Maio et al., 2023; Sanderson, 2021; Turakhia et al., 2021), visualization (Sanderson, 2022), and lineage identification (McBroome et al., 2024), an emerging body of research has been dedicated to efficient representations of data for genomes (Deorowicz et al., 2023) and phylogenetic trees such as phylo2vec (Chauve et al., 2025; Penn et al., 2024; Richman et al., 2025). Compared to the traditional Newick format which represents trees using strings of nested parentheses (Felsenstein, 2004), modern tree representations utilize integer vectors to define the tree topology traversal. This approach offers several advantages, including easier manipulation, increased memory efficiency, and applicability to machine learning.

Here, we present the latest release of `phylo2vec` (or Phylo2Vec), a high-performance software package for encoding, manipulating, and analysing binary phylogenetic trees. At its core, the package is based on the phylo2vec (Penn et al., 2024) representation of binary trees, and is designed to enable fast sampling and tree comparison. This release features a core implementation in Rust for improved performance and memory efficiency (Figure 1), with wrappers in R and Python (superseding the original release (Penn et al., 2024)), making it accessible to a broad audience in the bioinformatics community.

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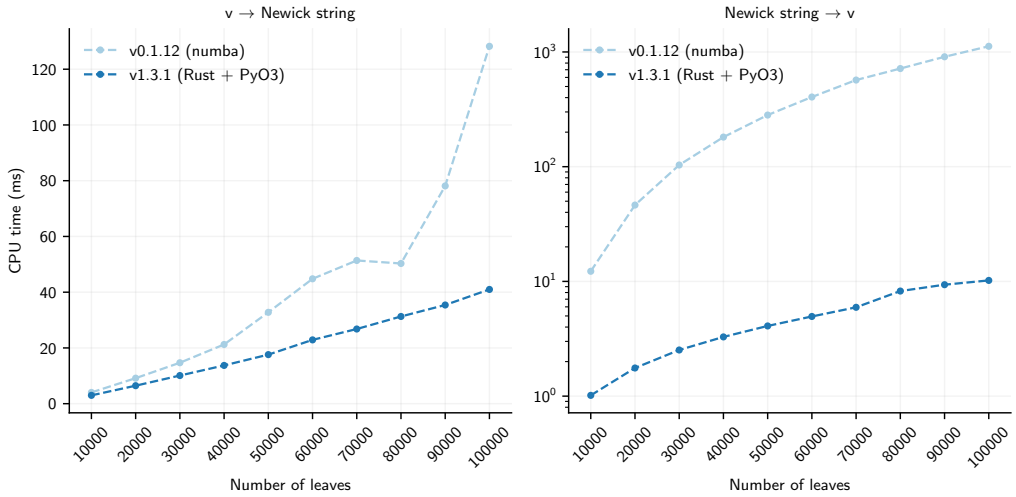

Figure 1: Benchmark times for converting a phylo2vec vector to a Newick string (left) and vice versa (right). Execution time was measured over at least 20 runs per size, comparing Python functions in the latest release (via Rust bindings with PyO3) against the previous release (Penn et al., 2024) based on `Numba` (Lam et al., 2015). Benchmarks ran on an on an AMD Ryzen Threadripper PRO 5995WX (64 cores, 7 GHz, 256 GB RAM).

Statement of need

The purpose of the `phylo2vec` library is threefold. First, it provides robust phylogenetic tree manipulation in Rust, complementing other efforts such as `light_phylogeny` (Duchemin et al., 2018) for reconciled phylogenies (Nakhleh, 2013), and `rust-bio` (Köster, 2016), which does not yet cover phylogenetics. Second, it complements existing libraries such as ape (Paradis & Schliep, 2019) in R, and `ete3` (Huerta-Cepas et al., 2016) and `DendroPy` (Moreno et al., 2024) in Python, by providing fast tree sampling, fast tree comparison and efficient tree data compression (Penn et al., 2024). Third, the phylo2vec representation offers a pathway to using new optimisation frameworks for phylogenetic inference. A notable example is GradME (Penn et al., 2023), a gradient descent-based algorithm which uses a continuous relaxation of the phylo2vec representation.

Features

The presented release of `phylo2vec` addresses optimisations limitations of (Penn et al., 2024) with $\mathcal{O}(n \log n)$ implementations for vector-to-Newick and Newick-to-vector conversions, leveraging Adelson-Velsky and Landis (AVL) trees (Adelson-Velsky & Landis, In Russian. English translation in Soviet Mathematics Doklady, 3:1259--1263, 1962) and Fenwick trees (Fenwick, 1994), respectively.

New features include supporting branch length annotations by extending the vector representation of size $n-1$ leaves to a matrix of size $(n-1) \times 3$ (Figure 2), leaf-level operations (pruning, placement, MRCA identification), fast cophenetic distance matrix calculation, and various optimisation schemes based on phylo2vec tree representations, notably hill-climbing (Penn et al., 2024) and GradME (Penn et al., 2023). Bayesian MCMC inference is enabled through a likelihood function that leverages tree representation similarities with `BEAGLE` (Ayres et al., 2012; Suchard & Rambaut, 2009). Finally, user-friendliness is enhanced with step-by-step demos of the inner workings of phylo2vec's representations and their conversion to common phylogenetic tree formats.

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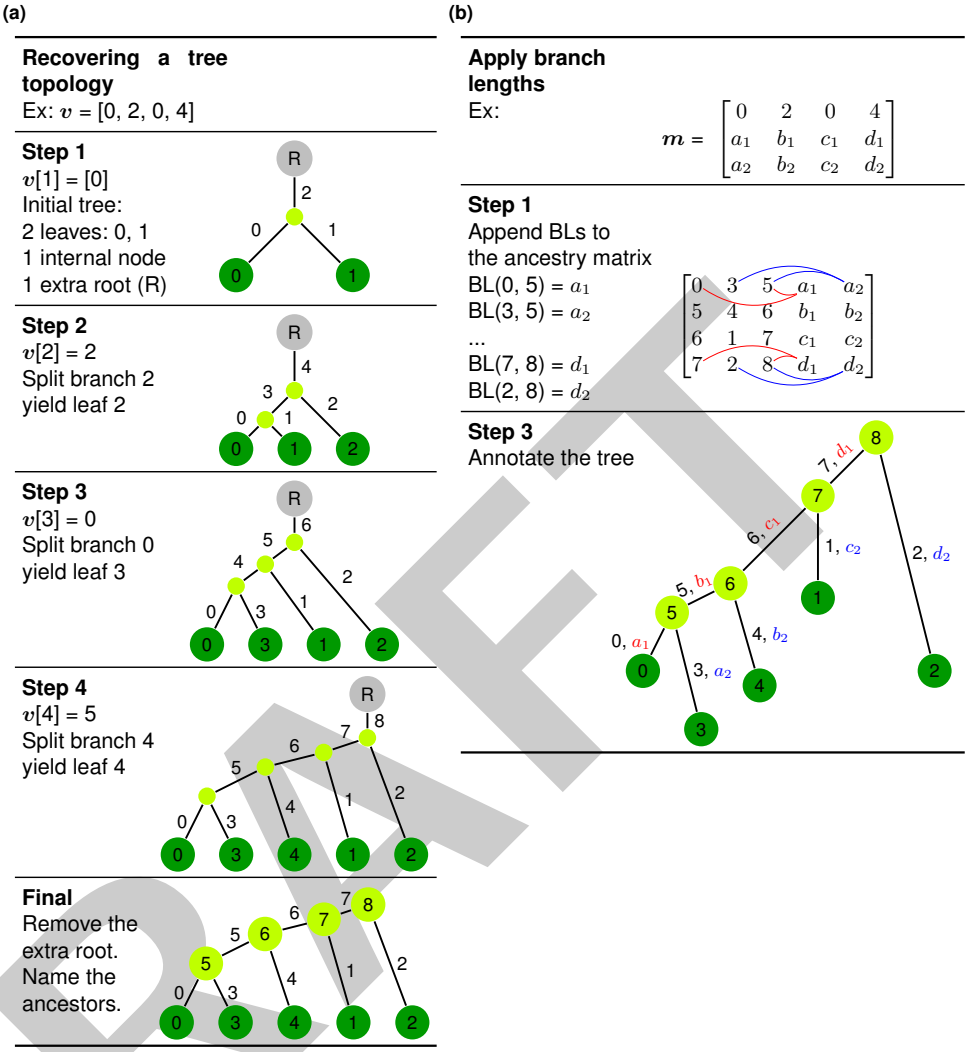

Figure 2: Recovering a tree from a phylo2vec vector: example for $v = [0, 2, 0, 4]$. (a) Leaf placement algorithm from (Penn et al., 2024). (b) Augmenting v into a matrix m with branch lengths. We use an intermediary ancestry matrix whereby each row encodes a cherry (two children + parent) with two extra columns for branch lengths. The node with the smallest descendant has the branch length in the second column, and the other the branch length in the third column.

Maintenance

A strong focus of this release is to support long-term maintenance through implementing recommended software practices into its project structure and development workflow. The project is structured with a Rust API containing core algorithms with language bindings to avoid tight coupling and enable easy language additions. Additionally, we have established a robust continuous integration (CI) pipeline using GitHub Actions, which features:

- Unit test frameworks for Rust (cargo), Python (pytest), and R (testthat (Wickham, 2011))
- Benchmarking on the Rust code (criterion) and its Python bindings (pytest-benchmark)

Lastly, to complement Jupyter Notebook demos, comprehensive documentation is provided using Jupyter Book and Read The Docs.

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